IGCSE 0607 Practice Questions — E1.9 — Combined Questions & Markscheme

Estimation

Total Questions: 18 — **Total Marks:** 90 Marks

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1 Consider the number 47.5381.

(a) Write 47.5381 correct to:

(i) 2 decimal places.

..... **[1]**

(ii) 2 significant figures.

..... **[1]**

(iii) the nearest integer.

..... **[1]**

2 Write the following numbers correct to the specified degree of accuracy.

(a) 782.51 correct to the nearest ten.

..... **[1]**

(b) 0.03561 correct to 2 significant figures.

..... **[1]**

(c) 4.098 correct to 2 decimal places.

..... **[1]**

3 **(a)** Write 48.975 correct to 2 decimal places.

..... **[1]**

(b) Write 0.003058 correct to 2 significant figures.

..... **[1]**

(c) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{7.84 \times 31.6}{0.385}$$

..... **[2]**

4 **(a)** Round the following numbers to the specified degree of accuracy.

(i) 34.567 correct to 2 decimal places.

..... **[1]**

(ii) 0.070381 correct to 2 significant figures.

..... **[1]**

(b) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{61.3 \times 3.92}{0.485}$$

..... **[2]**

5 **(a)** Write 0.04758 correct to 2 significant figures.

..... **[1]**

(b) Write 384.6 correct to the nearest 10.

..... **[1]**

(c) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{58.7 \times 4.12}{19.6 - 2.87}(0)$$

..... **[2]**

6 **(a)** Write 203.498 correct to

(i) 2 decimal places.

..... **[1]**

(ii) 3 significant figures.

..... **[1]**

(iii) the nearest ten.

..... **[1]**

(b) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{6.89 \times 41.2}{0.513}$$

..... **[2]**

7 By writing each number correct to 1 significant figure, estimate the value of

$$\frac{58.4 \times 3.09}{0.187}$$

..... **[3]**

8 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{7.84 \times 31.6}{0.381}$$

..... **[2]**

(b) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{98.6} + 19.3}{0.513^2}$$

..... **[3]**

9 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{98.5} \times 4.12^2}{0.187}$$

..... **[3]**

(b) A rectangular field has length 489 m and width 215 m.

Estimate the area of the field, in square metres, by rounding each length to 1 significant figure.

..... m^2 **[2]**

10 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{3.87 \times 10^6}}{0.0524 \times 21.3}$$

..... **[3]**

(b) Is your estimate in part (a) greater than or less than the exact value of the expression? Give a reason for your answer.

.....

..... **[2]**

11 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt[3]{8.25} \times 19.3^2}{0.485}$$

..... **[3]**

(b) The number of bacteria in an experiment is given by the formula

$$N = 4.12 \times 10^5 \times (1.95)^t$$

where t is the time in hours.
By writing each base number correct to 1 significant figure, estimate the number of bacteria when $t = 3$.
Give your answer in standard form.

$N \approx$ ..... **[3]**

12 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{98.5} \times 4.1^3}{0.192 \times 48.7}$$

You must show all your working.

..... **[3]**

(b) Given that $y = \dfrac{ax}{b-c}$, find an estimate for the value of y when
$a = 3.92$, $x = 0.0514$, $b = 1.08$ and $c = 0.812$.
Write all numbers correct to 1 significant figure in your calculation.

$y =$ ..... **[3]**

13 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{38.7 \times 5.12^2}{\sqrt[3]{996}}$$

..... **[3]**

(b) A solid metal cylinder has radius 3.89 cm and height 11.2 cm.

By writing each dimension correct to 1 significant figure, estimate the volume of the cylinder.
Take the value of π to be 3.

..... cm^3 **[3]**

(c) A train travels a distance of 4210 km at an average speed of 585 km/h.

By writing both the distance and the speed correct to 1 significant figure, estimate the time taken for the journey.
Give your answer in hours and minutes.

hours, minutes **[3]**

14 A rectangular field has length 125 m and width 74 m.

(a) Calculate the area of the field.
Give your answer correct to 2 significant figures.

..... m^2 **[2]**

(b) A different field is a square of side x metres.
The area of this square field is equal to the area of the rectangular field.
Calculate the value of x.
Give your answer correct to 1 decimal place.

$x =$ ..... **[3]**

15 A solid metal cone has radius $r = 5.82$ cm and height $h = 11.4$ cm.

(a) Calculate the volume of the cone.
Give your answer correct to 3 significant figures.
The volume, V, of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.

..... cm^3 **[2]**

(b) By writing each of the values 5.82, 11.4 and π correct to 1 significant figure, estimate the volume of the cone.
Show your working.

..... cm^3 **[2]**

(c) A worker packs these cones into boxes.
Each box holds exactly 12 cones.
The factory produces 850 cones a day.
Calculate the maximum number of boxes that can be completely filled each day.

..... **[2]**

16 **(a)** Calculate the value of

$$\frac{4.2^3 + \sqrt{18.5}}{0.75 \times 1.2}$$

Give your answer correct to 3 significant figures.

..... **[2]**

(b) A spherical water tank has a radius of 1.85 metres.

(i) Calculate the volume of the tank.
Give your answer correct to the nearest cubic metre.
The volume, V, of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.

..... m^3 **[2]**

(ii) The tank is initially empty.
Water is pumped into the tank at a rate of 12.5 litres per minute.
Calculate the time required to fill the tank completely.
Give your answer in hours and minutes, correct to the nearest minute.

hours, minutes **[3]**

17 **(a)** Calculate the value of

$$p = \frac{28.7^2 - 11.2^2}{\sqrt{98.5}}$$

(i) Write down all the figures on your calculator display.

$p =$ ..... **[1]**

(ii) Write your answer to part (a)(i) correct to 3 significant figures.

$p =$ ..... **[1]**

(b) By writing each number in the original expression correct to 1 significant figure, estimate the value of p.

$p \approx$ ..... **[2]**

18 A solid metal cylinder has a base radius of 4.15 cm and a height of 19.8 cm. The metal has a density of 8.13 grams per cubic centimetre.

(a) By writing each number correct to 1 significant figure, estimate the mass of the cylinder in grams.
Use $\pi \approx 3$ for your estimation.
Show your working clearly.

..... g **[3]**

(b) Calculate the actual mass of the cylinder.
Give your answer correct to 3 significant figures.

..... g **[3]**

1 **(a)** **(i)** 47.54 **[B1]**

(ii) 48 **[B1]**

(iii) 48 **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** 780 **[B1]**

(b) 0.036 **[B1]**

(c) 4.10 **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** 48.98 **[B1]**

(b) 0.0031 **[B1]**

(c) 600 **[A1]**

M1 for $\frac{8 \times 30}{0.4}$

[Back to Question] *[Total Marks: 4]*

4 **(a)** **(i)** 34.57 **[B1]**

(ii) 0.070 **[B1]**

(b) $\frac{60 \times 4}{0.5}$ **[M1]**

480 **[B1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** 0.048 **[B1]**

(b) 380 **[B1]**

(c) $\frac{60 \times 4}{20 - 3}$ or $\frac{240}{17}$ **[M1]**

14 or 14.1 (from $\frac{240}{17}$) or 15 (if 17 is estimated as 20) **[A1]**

[Back to Question] *[Total Marks: 4]*

6 **(a)** **(i)** 203.50 **[B1]**

(ii) 203 **[B1]**

(iii) 200 **[B1]**

(b) 560 **[A1]**

If 0 scored, M1 for $\frac{7 \times 40}{0.5}$ **[M1]**

[Back to Question] *[Total Marks: 5]*

7 $\frac{60 \times 3}{0.2}$ **[M1]**

$\frac{180}{0.2}$ **[M1]**

900 **[A1]**

[Back to Question] *[Total Marks: 3]*

8 (a) $\frac{8 \times 30}{0.4}$ **[M1]**

600 **[A1]**

(b) $\sqrt{100}$ or 10, and 20, and 0.5 **[M1]**

$\frac{10 + 20}{0.5^2}$ or $\frac{30}{0.25}$ **[M1]**

120 **[A1]**

[Back to Question] *[Total Marks: 5]*

9 (a) $\frac{\sqrt{100} \times 4^2}{0.2}$ **[M1]**

$\frac{10 \times 16}{0.2}$ or $\frac{160}{0.2}$ **[M1]**

800 **[B1]**

(b) 500×200 **[M1]**

100000 **[B1]**

[Back to Question] *[Total Marks: 5]*

10 (a) 2000 **[A3]**

If 0 scored, M1 for rounding the numbers to 4×10^6, 0.05 and 20 **[M1]**

M1 for evaluating the numerator to 2000 and the denominator to 1 **[M1]**

(b) Greater (than) **[B1]**

Valid reason, e.g. the numerator was rounded up and the denominator was rounded down. **[B1]**

[Back to Question] *[Total Marks: 5]*

11 (a) 1600 **[A1]**

M1 for $\frac{\sqrt[3]{8} \times 20^2}{0.5}$

M1 for 2×400 or 800 seen in numerator

(b) 3.2×10^6 **[A1]**

M1 for $4 \times 10^5 \times 2^3$

M1 for 32×10^5 or 3200000 seen

[Back to Question] *[Total Marks: 6]*

12 **(a)** $\sqrt{100} \times 4^3$ or 10×64 $(= 640)$ **[M1]**

0.2×50 $(= 10)$ **[M1]**

$\frac{10 \times 64}{10} = 64$ **[A1]**

(b) Uses $a = 4$, $x = 0.05$, $b = 1$, $c = 0.8$ **[M1]**

$\frac{4 \times 0.05}{1 - 0.8}$ or $\frac{0.2}{0.2}$ **[M1]**

1 **[A1]**

[Back to Question] *[Total Marks: 6]*

13 **(a)** 40, 5, and 1000 seen **[M1]**

$\frac{40 \times 5^2}{\sqrt[3]{1000}}$ or $\frac{40 \times 25}{10}$ **[M1]**

100 **[A1]**

(b) Radius ≈ 4 and Height ≈ 10 **[B1]**

Volume $= 3 \times 4^2 \times 10$ **[M1]**

480 **[A1]**

(c) Distance ≈ 4000 and Speed ≈ 600 **[B1]**

$\frac{4000}{600}$ or $\frac{20}{3}$ or 6.66... hours **[M1]**

6 hours and 40 minutes **[A1]**

[Back to Question] *[Total Marks: 9]*

14 **(a)** $125 \times 74 = 9250$ **[M1]**

9300 **[A1]**

(b) $x^2 = 9250$ **[M1]**

$x = \sqrt{9250}$ (96.176...) **[M1]**

96.2 **[A1]**

[Back to Question] *[Total Marks: 5]*

15 **(a)** 404 **[A2]**

If 0 scored, M1 for $\frac{1}{3} \times \pi \times 5.82^2 \times 11.4$ **[M1]**

(b) 360 **[A2]**

If 0 scored, M1 for $\frac{1}{3} \times 3 \times 6^2 \times 10$ **[M1]**

(c) 70 **[A2]**

If 0 scored, M1 for $850 \div 12$ or 70.83... seen **[M1]**

[Back to Question] *[Total Marks: 6]*

16 (a) 87.1 **[B2]**

M1 for 87.098... seen or $\frac{78.38...}{0.9}$

(b) (i) 27 **[A1]**

M1 for $\frac{4}{3} \times \pi \times 1.85^3$ or 26.5... seen

(ii) 35 hours 22 minutes **[A1]**

M1 for their exact volume (or 27) $\times 1000$ (e.g. 26522 litres)

M1 for $\frac{\text{their } 26522}{12.5}$ $(= 2121.76$ mins$)$

[Back to Question] *[Total Marks: 7]*

17 (a) (i) 70.35465... **[B1]**

Accept an answer containing at least 5 decimal places.

(ii) 70.4 **[B1]**

(b) $\frac{30^2 - 10^2}{\sqrt{100}}$ or $\frac{900 - 100}{10}$ **[M1]**

80 **[B1]**

[Back to Question] *[Total Marks: 4]*

18 (a) $r \approx 4$, $h \approx 20$, density ≈ 8 **[M1]**

Estimated Mass $= 3 \times 4^2 \times 20 \times 8$ **[M1]**

7680 **[A1]**

(b) Volume $= \pi \times (4.15)^2 \times 19.8 = 1071.3\ldots$ **[M1]**

Mass $= 1071.3\cdots \times 8.13 = 8710.19\ldots$ **[M1]**

8710 **[A1]**

[Back to Question] *[Total Marks: 6]*